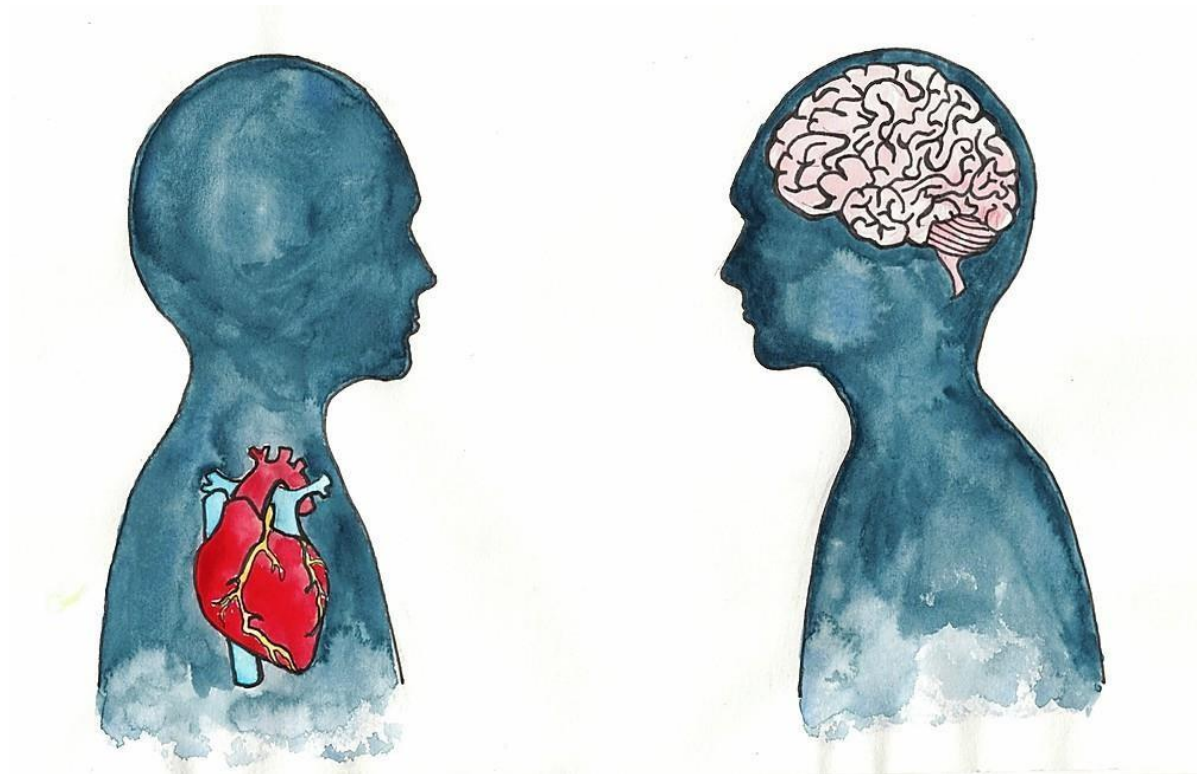


COMPREHENSIVE PHYSICAL AND MENTAL WELL-BEING OF SLIIT CITY UNI STUDENTS

Probability and Statistics

SCU



Abstract

Health is one of the most important yet one of the most ignored problems in the modern world. The physical and mental well-being of students in a university is important as the student will one day grow up to be the responsible leaders of society. This report contains statistical data on Comprehensive physical and mental well-being of SCU students and the conclusions that have been reached by studying the data gathered through a survey deployed among students in SCU from foundation level to third year level. Furthermore, the report includes techniques used to gather data, strategies used to group and analyze descriptive and numerical data, stages in analyzing data to achieve conclusions in order. The purpose of the report is to achieve useful conclusions on current physical and mental well-being of students in SCU using the sample data gathered.

Declaration

I, _____, a student of Sliit City University currently enrolled in the module probability and statistics declare this report on “Comprehensive physical and mental well-being of Sliit City Uni students” an original work created for continuous assessments for the module Probability and Statistics. The data that have been used in this report are not original, have been appropriately acknowledged with utmost respect to their original owners. Also, I assure confidentiality and anonymity of data gathered from students for this research and they will not be used outside this survey. Furthermore, I alone bear responsibility for impacts and outcomes that could be produced by this report.

([DATE] [Month] [YEAR])

Acknowledgements

I offer my utmost gratitude to everyone who has helped me carry out the “Comprehensive physical and mental well-being of Sliit City Uni students” survey successfully and prepare this report. Specially I would like to offer my utmost gratitude to Mr. Chamith Jayasinghe, the lecturer in charge of Probability and Statistics module for providing me with all the knowledge necessary to carry out this research successfully from the beginning and all the students of Sliit City Uni who participated in this survey by providing me with accurate and correct data to make this research successful. Carrying out this research would have been impossible without all the help and support I received from every individual.

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Chapter 1 – Introduction

1.1 Background

Health. Health is one of the most important things for any and all living beings. Health is described as **“The state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity”** (WHO, 1948) by the WHO*. As the definition suggests, health is not just absence of diseases. To further break it down, even someone who suffers from a disease can still be healthy. Mainly health or overall well-being can be divided into two parts.

1. Physical Health
2. Mental Health

These aspects of health are dependent, meaning well-being of one aspect can result in the well-being of the other as well as the lack of one aspect can affect negatively on the other aspect.

Physical health

Physical health can be referred to as the normal functioning of the body. There are many factors that can determine a person’s physical health. Physical activity, nutrients and diet, rest and sleep, medical self-care are a few of them.

Mental health

Mental health can be described as the state of mental well-being that enables people to cope with the stresses of life, realize their abilities, and contribute to their community. In short, mental health is an individual’s capacity to feel, think and act in ways to achieve a better quality of life. Social, psychological, and emotional needs are included in mental health.

The well-being of these 2 aspects of health ultimately decides the overall well-being of a person.

To a student, physical and mental well-being is a crucial factor that decides their overall performance in their field of study. To study is to understand. To understand, one must have a mind that lets them freely consume information and think through them. Also, they should have a physical body that does not hold their mind back from utilizing its total potential. Due to these reasons, Physical and mental well-being of students is an important matter that should be paid attention to.

1.2 Problem Identification

Maintaining a good health is very important for everyone as it helps them live longer, be more productive, be more confident, lower the risk of chronic diseases, and have good relationships with people. The problem with Physical and mental well-being all around the world is that it is not recognized as much as it should be. The majority of people do not realize how important it is to be healthy. With the current industrial society being competitive, people have forgotten the importance of health. The priority for health in people's lives is gradually getting lower.

The less attention paid to health problem is more critical in students due to their significant importance to the future of the country and the world. Lack of attention to health should not get in their way as it can negatively affect the process of learning which will ultimately lead to a decrease in academic performance. The main problem this research attempts to answer is whether the level of physical and mental well-being of Sliit City Uni student is in a positive range and does it require more attention than the amount of attention given to it by students.

1.3 Significance of research

This research evaluates the levels of physical and mental well-being of Sliit city Uni students. By carrying out this research successfully, it is possible to identify where the level of physical and mental well-being lies in current Sliit City Uni students. Furthermore, carrying out this research will be helpful in identifying how academic performance affects mental health of students and how their mental health affects academic performance as well. This research concludes whether it is required for students to pay more attention to their health or is the current amount of attention given to their own health by students is sufficient. Success of This research will be the first step towards attempting to improve students' overall health. New information extracted from this research can be utilized to further take decisions regarding what affects students' health and to reduce what affects them negatively. The major importance of this research is that it generates new statistical information that can lead to new researches regarding overall health of Sliit City Uni Students.

1.4 Objectives of the study

The main purpose of this study is to figure out the levels of overall mental and physical well-being of students at Sliit City Uni. All the objectives this research attempts to complete are presented below in order.

- Identifying how active students are on a weekly basis.
- Identifying how many hours of sleep students get on average.
- Identifying students' body weight and height distribution.
- Identifying if the student's mental health affects their academic performance in a negative way.
- Identifying if their academic performance affects their mental health in a negative way.
- Identifying if students have habits that could affect their health negatively.
- Analyzing where the level of overall physical and mental well-being of Sliit City Uni students lies using statistical techniques.
- Arriving at conclusions using the data analyzed and answering the questions of "Should students pay more attention to their physical and mental health?" and "Is the overall mental and physical well-being of students in Sliit City Uni good?"

1.5 Chapter Framework

This chapter is an introduction to the research called "Comprehensive physical and mental well-being of Sliit City Uni students" which is study carried out to answer the questions of "Where does overall physical and mental health level of Sliit City Uni Students lies?" and "Do the students need to pay more attention to their health than they are already paying?"

- 1.1. Background** – Describes what health is and how important it is for people, including students, to pay attention to their health.
- 1.2. Problem Identification** – Addresses the problem of people not paying enough attention to their health. Addresses how the problem is more critical when it comes to students.
- 1.3. Significance of research** – Describes the importance of carrying out the research including the new statistical information it can generate.
- 1.4. Objectives of the study** – States the purpose of carrying out the research and states research objectives in order.
- 1.5. Chapter framework** – A summary of the first chapter.

Chapter 2 - Literature Review

It is necessary to investigate previous researches related to mental and physical well-being of university students carried out by various researchers around the world to identify whether this research on mental and physical well-being of Sliit City Uni students results in a conclusion different than the previous researches or not. It is important to carry out an investigation into what has already been researched to find out what is already known and what is not.

There are many researches conducted on mental and physical well-being of university students around the world using various approaches. A study by *Laska et al. (2011)* on physical activity behavior among college students concludes that many students fail to meet recommended physical activity guidelines. A meta-analysis by *Kandola et al. (2019)* explored the association between physical activity and mental well-being among university students, highlighting the positive impact of exercise on reducing symptoms of depression and anxiety. Research by *Nelson et al. (2010)* investigated dietary habits among college students and identified a high consumption of unhealthy foods and low intake of fruits and vegetables. A study by *Eisenberg et al. (2013)* surveyed college students to assess the prevalence of mental health disorders and found high rates of stress, anxiety, and depression. Research by *Hefner and Eisenberg (2009)* examined the role of social support networks in buffering the impact of stress on student well-being.

The above-mentioned studies study deeper into a specific sector of overall well-being of university students. What differs them from the study of comprehensive mental and physical well-being of Sliit City Uni students is the population the research is carried on and the topic which is not specifically focused on a single sector of health but overall health. This study will bring forth new data on the specific population, which is all the current Sliit City Uni students.

Chapter 3 - Theory and Methodology

3.1 Research Design

This study on comprehensive physical and mental well-being of Sliit City Uni students is mainly focused on gathering primary data that can be used to perform statistical calculation and draw out conclusions on overall health of Sliit city Uni students from them. The research is designed in a way that it is easy to gather accurate data from students.

As the sampling method to choose a sample from the selected population convenience sampling method which is a non-probability sampling method is used. Reasons for this sampling method to be chosen over others are the simplicity of the sampling method, having to gather a sufficient amount of data from the sample to represent the population within a short period of time, and low response rate. However, using a non- probability sampling method may result in a sample that does not properly represent the targeted population.

The survey to gather data has been carried out as an online questionnaire that mainly consists of closed-ended questions. An online questionnaire as the survey method was selected due to its ability to collect data from a large sample within a short period of time.

To analyze data gathered by the survey both categorical data analyzing methods and numerical data analyzing methods have been used. Frequency tables and pie charts are mainly used to describe categorical data. To describe numerical data, mean, median, mode, standard deviation, variance, range, and bar graphs have been used.

To make conclusions on the population using the sample hypothesis technique with one sample test has been used.

Chi-squared test is used to find independency of categorical variables to make it easier to make conclusions from those data.

3.2 Data Collection Method

As the data collection method, a survey has been deployed using a google form questionnaire targeting students at Sliit City Uni. The questionnaire mainly consists of closed-ended questions that make it easier to analyze answers gathered from them and are helpful in extracting necessary quantitative information needed to carry out

the research successfully. However, there is a lack of open-ended questions in the questionnaire since qualitative data is not mainly focused on when carrying out the research. Every student currently studying at Sliit City Uni were allowed to participate in this survey to collect data on their overall health.

3.3 Structure of the Questionnaire

The questionnaire titled “Comprehensive Physical and Mental Well-being of Sliit City Uni Students” deployed using google forms consisted of 15 questions that were necessary to collect data on students’ overall health. Participants of the survey were required to answer the questions from 1 to 14 while they were given the freedom to choose whether to answer the 15th question or not.

- The first 3 questions were focused on collecting data necessary to categorize information.
 1. Year of study
 2. Gender
 3. Age group
- Questions from 4 to 9 are to gather information on the physical well-being of students.
 4. How much do you weigh?
 5. What is your height?
 6. How many hours do you sleep on average?
 7. Do you drink enough water on average?
 8. How often do you usually engage in physical activities on a weekly basis?
 9. How physically active are you throughout the day on average?
- Questions from 10 to 14 are focused on gathering information on the mental well-being of students.
 10. How do you feel about yourself?
 11. Does your mental health have a negative impact on your academic performance?
 12. Does your academic performance have a negative impact on your health?
 13. Do you have any unhealthy habits that could negatively affect your mental or physical well-being?
 14. Do you often feel mentally exhausted?

- The last open-ended question was reserved for students to describe more on their overall physical and mental health if the questions asked were not enough for them to describe their overall health.

15. Any extra comments?

3.4 Preliminary Data Analysis

3.4.1 Descriptive Analysis

Descriptive analysis under preliminary data analysis is carried out to identify characteristics of data, patterns of data distributions, and mainly to discover anomalies in data collected before going deeper into analyzing all the data and making conclusions.

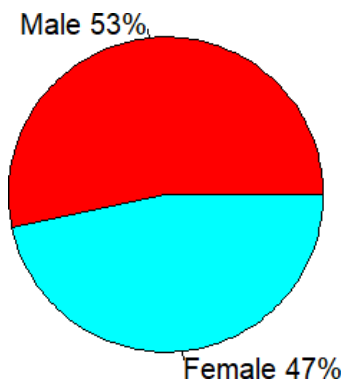
Categorical and numerical data analysis techniques will be used in descriptive analysis. As categorical techniques contingency tables, frequency tables and pie charts will be used while as numerical techniques mean, median, mode, range, variance, standard deviation and bar graphs will be used.

3.4.2 Demographic Data Analysis

Demographic analysis is conducted to find out how the demographic data are distributed throughout the sample that is used to carry out this survey. Demographic data can be described as attributes that can define the population. It describes characteristics in the population. For this survey, Gender, Age, Year of study can be considered as demographic attributes.

Distribution of Gender

The attribute “Gender” contains 2 possible values. They are “Male” and “Female”. Below pie chart of gender distribution explains how the male and female attributes are distributed throughout the sample of 30 students.



Male	Female
16	14

Table 3.4.2.1: Gender table

Figure 3.4.2.1: Pie Chart Gender

Out of the population of 30 total of 16 male students and 14 female students have participated in the survey. As it can be seen the majority of the participants are male students with a difference of 2 (6%).

Distribution of Age

In the survey deployed age was considered as a categorical variable that was categorized into 4 levels.

1. Less than 18
2. 18-24
3. 25-30
4. Above 30

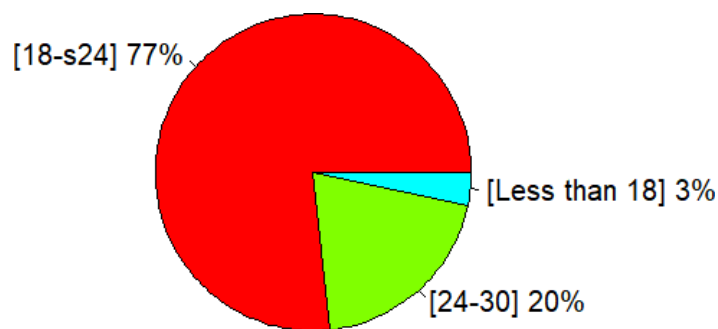


Figure 3.4.2.2: pie chart Age

Less than 18	18-24	25-30	Above 30
1	23	6	0

Table 3.4.2.2: Table Age

As it is represented, there is no slice in the pie chart for the age group “Above 30”. This has occurred because the number of participants from the age group above 30 was 0 in the survey. As it is demonstrated in the graph, the majority of students who have participated in the survey are 18 to 24 years old students.

Distribution of Year of Study

Year of study is a categorical variable with 4 possible answers. This attribute is important when comparing overall health levels of students from each year of study.

1. Foundation
2. First Year
3. Second Year
4. Third Year

This variable can also be considered as an ordinal variable because it can be ranked from foundation to 3rd year.

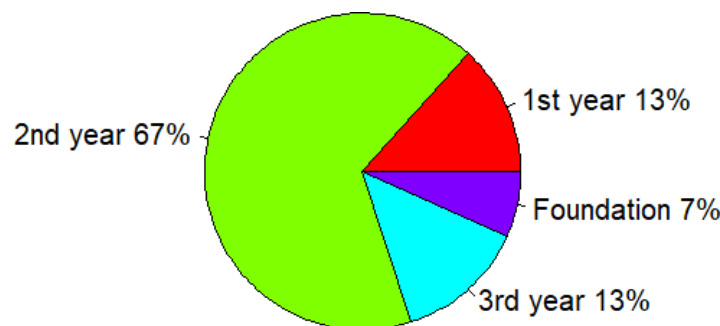


Figure 3.4.2.3: Pie chart Year

Foundation	1 st Year	2 nd Year	3 rd Year
2	4	20	4

Table 3.4.2.3: Table Year

As it can be seen the majority of students participated in this survey are from 2nd year. This might cause errors in conclusions since this population does not represent the sample accurately. This happened because the sample size taken was not enough to represent the population and the sample method used being a non-probability method (convenience sampling).

3.5 Descriptive data analysis

Under this chapter all the categorical and numerical attributes used in this survey will be described, analyzed and summarized. Furthermore, inferential statistic methods using the data collected will be presented in this chapter.

Attributes used in the survey.

Numerical	Categorical
• Weight • Height • BMI	• Year of study • Gender • Age • Sleep duration. • Daily water intake • Weekly exercise frequency • Activity level • Academic performance on mental health • Mental health on academic performance • Unhealthy habits • Mental exhaustion • Self-feeling

Table 4.1.1: Data categories

Categorical Attributes

Gender, Age, Year of Study will not be discussed under this chapter since they were already described in chapter 3.4.2. Demographic data analysis

Sleep Duration

This attribute is used to collect data on how many hours per day on average students at Sliit City Uni spend sleeping. This attribute can be used to make decisions on both physical and mental health of students since sleep is important to normal functioning of the human mind and body. It is recommended for a person to get 6-9 hours of sleep per day. Getting less sleep than 6 hours could have negative effects on people's lives. It makes people less productive and can have serious effects if it's continued for an extended period of time. Also sleeping more than they should, could also affect

people's health negatively. There were 3 possible answers students were given to choose from.

01. Less than 4 hours
02. 5-8 hours
03. More than 8 hours



Figure 4.1.1: Pie chart Sleep

Less than 4h	5h-8h	More than 8h
3	24	3

Table 4.1.2: Table sleep

80% of the total participants, which represent 24 students, regularly sleep 5 hours to 8 hours according to how the data are distributed in the pie chart.

Daily water intake

This attribute describes how much water a student takes regularly. Water intake is directly related to the physical health of people. Drinking enough water is compulsory for normal functioning of the human body. Drinking less water would cause serious health issues such as kidney stones and organ failures. This attribute is separated into 3 categories. There was an option given for students to choose in case they are not aware of their daily water intake.

1. 2.5 Liters – 4 Liters
2. Less than 2.5 Liters
3. Unaware.

Unaware	2.5l-4l	Less than 2.5L
7	17	6

Table 4.1.3: Water intake table

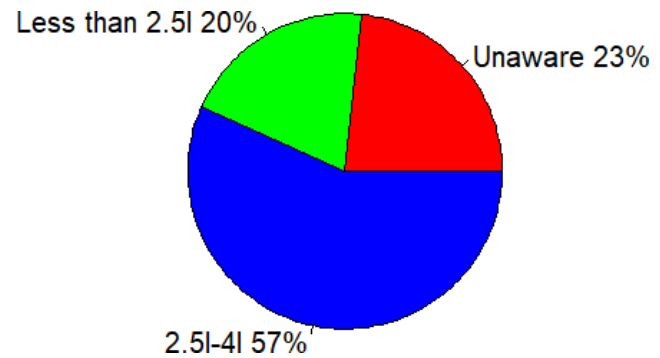


Figure 4.1.2: Pie chart Water intake

The majority of participants drink 2.5 to 4 liters of water per day regularly.

Weekly exercise frequency

This is another categorical variable which will be used to make decisions on students' physical health. This variable is used to collect data on how frequently students engage in physical exercises on a weekly basis. It is recommended to engage in physical activities at least 2 days per week. This variable contains 5 different possible values.

- 01. Once a week
- 02. 2-4 Times
- 03. More than 4 times
- 04. Rarely
- 05. Never

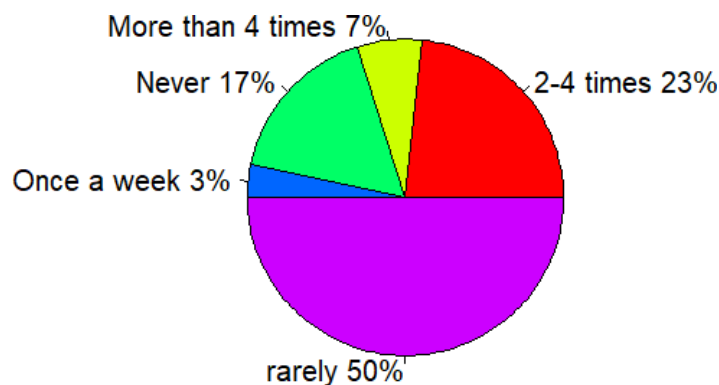


Figure 4.1.3: Pie Chart Weekly exercises

Never	rarely	Once a week	2-4 times	More than 4 times
5	15	1	7	2

Table 4.1.4: Table weekly exercises

An exact half, which is 15 of the sample of 30 rarely exercise on a weekly basis. The number of students who exercise on a weekly basis is 9, which is considerably low.

Activity Level

Activity level represents how active students are throughout the day. This variable will be used in determining physical health levels of students. Activity levels are separated into 4 levels or categories.

- 01. Very active
- 02. Active
- 03. Less active
- 04. Not active

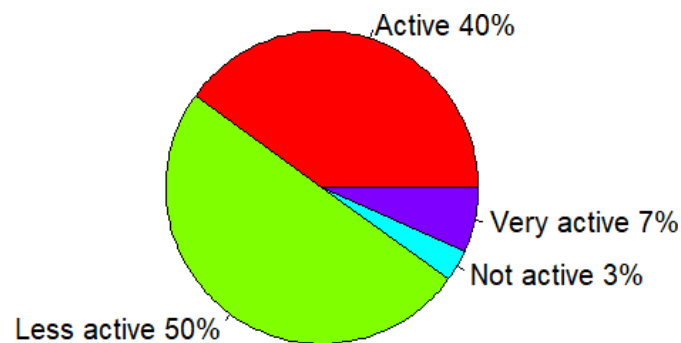


Figure 4.1.4: Pie chart Activity Level

Very Active	Active	Less Active	Not Active
2	12	15	1

Table 4.1.5: Table Activity Level

Half of the total sample of 30 15 students states they are less active throughout the day on average.

Academic performance on mental health

This variable is used to identify whether the academic performance of students can have a negative impact on their mental health. While this may be true for some students it might not be for others. There were 4 possible answers students were provided to choose from.

1. It has a negative impact.
2. It doesn't have a negative impact.
3. It sometimes has a negative impact.
4. Unaware

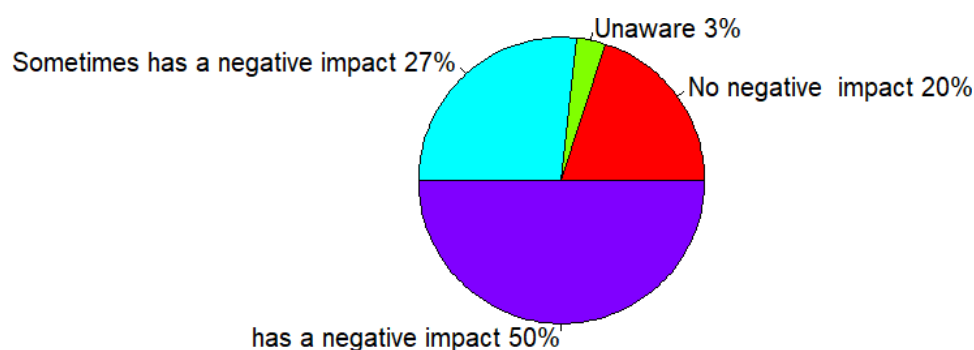


Figure 4.1.5: Pie chart Academic performance on mental health

Unaware	Has a negative Impact	No negative Impact	Sometimes it has a negative impact
1	15	6	8

Table 4.1.6: Table Academic performance on mental health

As the chart demonstrates, 50%, or 15 of all the students that participated in the survey states that they experience a negative impact on their mental health caused by their academic performance.

Mental health on academic performance

Mental health on academic performance describes the opposite of what the previous categorical attribute described. Instead of negative impact on mental health caused by academic performance of students, here we discuss how the students' Mental health impacts negatively in their academic performance. This attribute is also categorized into 4 parts.

1. It has a negative impact.
2. It doesn't have a negative impact.
3. It sometimes has a negative impact.
4. Unaware

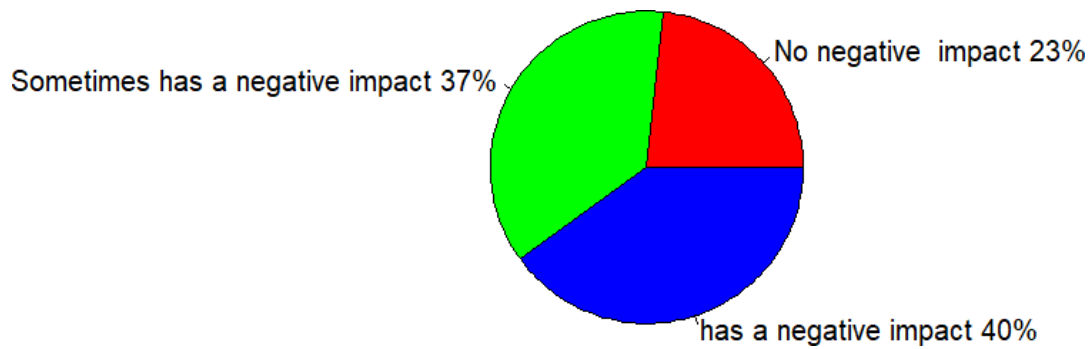


Figure 4.1.6: Pie chart Mental health on academic performance

Unaware	Has a negative Impact	No negative Impact	Sometimes it has a negative impact
0	12	7	11

Table 4.1.7: Table Mental health on Academic performance

As it is represented in the pie chart no student has chosen the category “Unaware” for the question “Does your Mental health negatively affect your Academic performance?”. The majority of the students who participated in the survey state that they do experience a negative impact on their academic performance because of their mental health.

Unhealthy habits

The unhealthy habits attribute focuses on both physical and mental health aspects as people can have unhealthy habits that could affect either of the 2 aspects of health or both. Unhealthy habits attribute was categorized into 3 parts for students to choose one of the three categories.

1. Have unhealthy habits.
2. Does not have unhealthy habits.
3. Unaware



Figure 4.1.7: Pie chart Unhealthy habits

Unaware	Does have unhealthy habits	Doesn't have unhealthy habits
7	13	10

Table 4.1.8: Table Unhealthy habits

The pie chart demonstrates that the majority of students who participated in the survey agree that they have unhealthy habits that could have a negative impact on their physical or mental health.

Mental Exhaustion

This is an attribute used in the survey to collect data on whether the students taken for the sample feel mentally exhausted frequently or not. This attribute can also be used when drawing conclusions on students' mental health. This variable is categorized into 3 parts.

1. Often mentally exhausted
2. Not often mentally exhausted.
3. Never feel mentally exhausted.

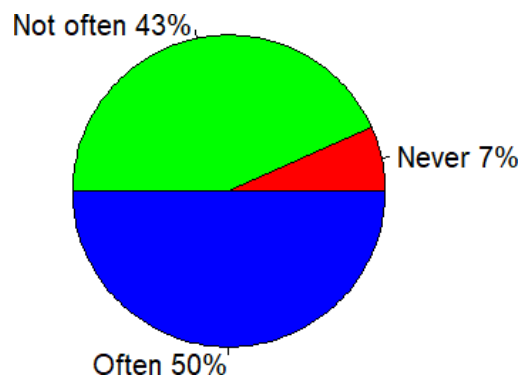


Figure 4.1.8: Pie chart Mental Exhaustion

Often mentally exhausted	Not often	Never
15	13	2

Table 4.1.9: Table Mental Exhaustion

As the pie chart demonstrates, the majority of students who participated in the survey feel mentally exhausted often.

Self-feeling

The attribute “Self-feeling” was included in the survey to extract data from students on whether they have a positive or a negative feeling about themselves. This attribute is going to be important and will be considered when making conclusions on students’ mental health levels. This attribute is categorized into 3 parts.

1. Positive feeling of self
2. Negative feeling of self
3. Neutral.

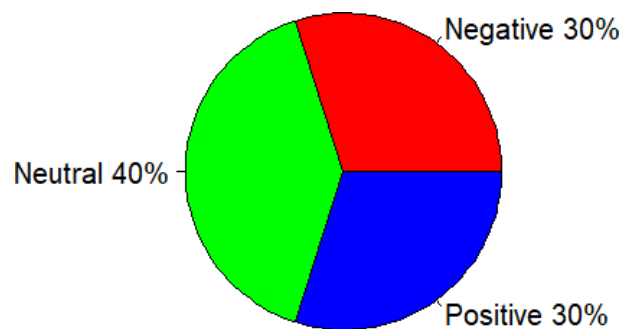


Figure 4.1.9: Pie chart “Self-feeling”

Positive	Negative	Neutral
9	9	12

Table 4.1.10: Table “Self-feeling”

The majority of students state that they have a neutral feeling about themselves, which means that they both equally feel negatively and positively about themselves.

Numerical Attributes

*Age which was previously considered as a categorical variable in this survey. It will be considered as numerical variable in this section. It's required for the variable age to be both numerical and categorical in this research. *

Sample will be divided into 2 parts as male and female in numerical attribute analysis in Height, Weight, and BMI.

Methods used in describing numerical data.

n = sample size.

i = position of value (starting from 1)

1. Sample arithmetic Mean – Average numerical value of data distribution (expected values). Summation of all values divided by the sample size.

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

Figure 4.1.10: Mean

2. Mode - It is the most frequent numerical value in the data distribution.
3. Median – Middle value of an ordered numerical data distribution where 50% of values are greater than this and 50% of values are less than this (array starts with 1). When the sample size is odd median is simply considered as the value in middle position. However, it becomes slightly different when it comes to an even size sample size.

- i. Finding the position (sample size is odd)

$$\frac{n+1}{2}$$

Figure 4.1.11: Median (position)

- ii. Median (Sample size is even)

- i. Position should be obtained. (x = position value obtained)

- ii. Mode = $x \text{ position value} + ((x+0.5) \text{ position value} - x \text{ position value})/2$
4. Range – It's the difference between the highest value of the distribution and the lowest value.

Range = **Highest value – Lowest value**

5. Variance – It's the measurement of the spread between numbers in a data set. It represents how the values are spread from the mean.

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

Figure 4.1.12: Variance

6. Standard Deviation – It is the measure of how dispersed the data is in relation to the mean.

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

Figure 4.1.13: Standard deviation.

Age

This attribute was previously considered as a categorical attribute. However, to carry out the research successfully it is necessary for the attribute Age to be both categorical and numerical.

Since it is a categorical variable with different categories, the middle value of each category will numerically represent each category and the data distribution.

Age category	Frequency	value
Less than 18	1	17
18-24	23	21
25-30	6	28
Above 30	0	31

Table 4.1.11: Table Numerical Age

[illegible]

Table 4.1.12: Table Age values

Height ordered array – 149.00, 152.00, 153.00, 153.00, 153.00, 155.00, 155.44, 157.00, 157.48, 159.00, 162.00, 163.00, 165.00, 165.10, 169.00, 170.00, 176.00, 178.00, 179.00, 180.00, 180.00, 180.34, 181.00, 182.00, 182.00, 186.00, 190.00, 190.00, 195.58, 198.00



Mean	Mode	Median	Range	Standard deviation	Variance
170.5313	153	169.5	49 (198–149)	14.45747	209.0185

Table 4.1.13: Table Height values

Weight

Weight is a numerical variable that is crucial in deciding physical health levels of students who participated in this survey. Weight and height together with Age calculate the BMI which is a standard measure in identifying physical fitness and health. This variable represents weight in Kilograms.

Weight ordered array – 35, 38, 41, 46, 47, 48, 49, 50, 51, 52, 53, 60, 60, 60, 63, 65, 72, 73, 74, 75, 76, 76, 78, 78, 80, 80, 90, 95, 100.

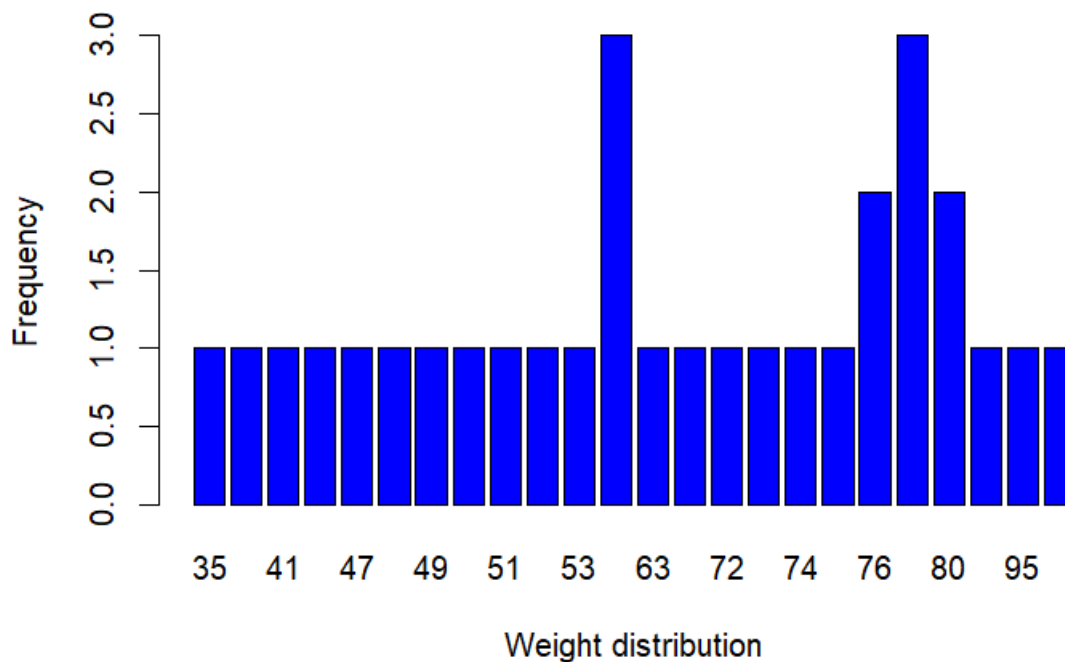


Figure 4.1.15: Bar graph of weight

Mean	Mode	Median	Range	Standard deviation	Variance
64.76667	60-78	64	65 (100-35)	17.10955	292.7368

Table 4.1.14: Table Weight values

BMI

$$BMI = \frac{Weight (kg)}{\{Height (m)\}^2}$$

Figure 4.1.16: BMI formula.

BMI is a standard measure used to category weight classes based on how an individual's weight is distributed with their height and age. Data for this variable was calculated and was not entered by participants of the survey.

BMI ordered array - 14.2, 15.0, 17.3, 17.3, 17.9, 18.1, 18.5, 19.4, 19.6, 19.7, 19.9, 20.0, 20.1, 21.6, 22.0, 22.2, 22.5, 22.5, 23.0, 23.1, 23.7, 23.8, 25.0, 25.2, 25.6, 26.3, 26.9, 28.2, 30.2, 31.2.

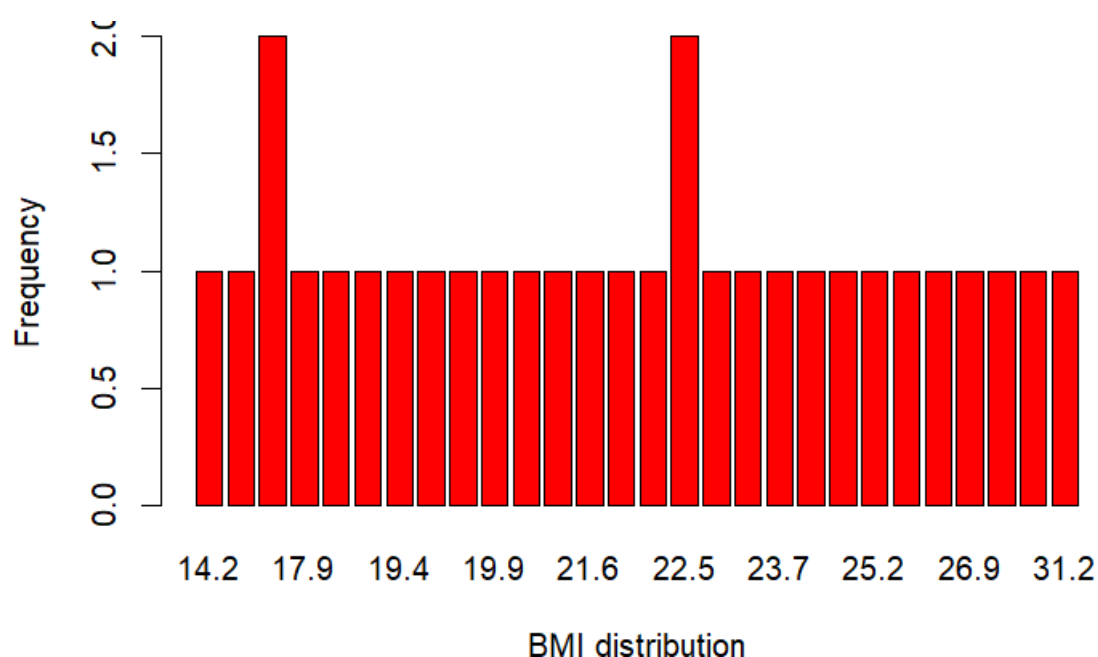


Figure 4.1.17: BMI distribution.

Mean	Mode	Median	Range	Standard deviation	Variance
22	17.3 – 22.5	22.1	17 (31.2-14.2)	4.161316	17.31655

Table 4.1.15: Table BMI

Female

14.2, 15.0, 17.3, 18.5, 19.6, 19.7, 19.9, 20.0, 22.0, 22.5, 25.6, 26.9, 28.2, 31.2

Mean	Mode	Median	Range	Standard deviation	Variance
21.47124	14.2	19.95	17 (31.2-14.2)	4.961611	24.61758

Table 4.1.16: Table BMI values female

Male

17.3, 17.9, 18.1, 19.4, 20.1, 21.6, 22.2, 22.5, 23.0, 23.1, 23.7, 23.8, 25.0, 25.2, 26.3, 30.2

Mean	Mode	Median	Range	Standard deviation	Variance
22.4525	17.3	22.75	12.0 (30.2 -17.3)	3.413869	11.6545

Table 4.1.17: Table BMI values male

Chapter 4 – Results

4.1 Inferential Data Analysis

Hypothesis Testing

t-distribution is used.

Level of significance = 0.1 (higher than usual since sample doesn't represent the population well)

Reject H_0 if P-value < level of significance.

18-24 age group is considered as the population parameter in this scenario. Healthy BMI

range of females in age group 18-24 = 18.5 - 24.9

Healthy BMI range of male in age group 18-24 = 18.5 - 24.

- Below the recommended BMI will be considered underweight and over the limit will be considered as overweight

02. Are Sliit City Uni students underweight?

Null hypothesis – Sliit City Uni students are not underweighted. H_0 :

mean BMI ≥ 18.5

Alternative hypothesis – Sliit City Uni students are underweighted. H_1 : mean

BMI < 18.5

Female

```
> orderbmi <-sort(subset(weightData$BMI,weightData$Gender == "Female"))
> resultFemale <- t.test(orderbmi,mu=18.5,conf.level=0.9,alternative = 'less')
> resultFemale

One Sample t-test

data: orderbmi
t = 2.2408, df = 13, p-value = 0.9784
alternative hypothesis: true mean is less than 18.5
90 percent confidence interval:
 -Inf 23.26182
sample estimates:
mean of x
 21.47143

> |
```

Figure 4.1.18: Hypothesis testing: Underweight: Female

The data obtained from the t-test indicates,

- P-value of Null hypothesis being true = $1 - 0.9784$
= 0.0216

Since the level of significance is 0.1 p-value obtained from the test is less than the level of significance.

p-value < Level of significance = $0.0216 < 0.1$ Hence

H_0 can be rejected.

Conclusion – Female students of the Sliit City Uni are underweight.

Male

```
> orderbmi <-sort(subset(weightData$BMI,weightData$Gender == "Male"))
> resultmale <- t.test(orderbmi,mu=18.5,conf.level=0.9,alternative = 'less')
> resultmale

One Sample t-test

data: orderbmi
t = 4.6428, df = 15, p-value = 0.9998
alternative hypothesis: true mean is less than 18.5
90 percent confidence interval:
 -Inf 23.60666
sample estimates:
mean of x
 22.4625

> |
```

Figure 4.1.19: Hypothesis testing: Underweight: Male

The data obtained from the t-test indicates,

- P-value of Null hypothesis being true = $1 - 0.9998$
= 0.0002

Since the level of significance is 0.1 p-value obtained from the test is less than the level of significance.

$$p\text{-value} < \text{Level of significance} = 0.0002 < 0.1$$

Hence H_0 can be rejected.

Conclusion – Male students of the Sliit City Uni are underweight.

Hypothesis tests for both male and female students resulted in the same way. In both hypothesis tests null hypothesis was rejected, concluding students at Sliit City Uni are underweight.

Chi squared testing (Test for independence).

Level of significance = 0.1.

Chi-squared approximation might be inaccurate because the sample size is low Reject if,

1. $P\text{-value} < \text{level of significance}$.
2. Critical value $<$ Chi squared value.

03. Are Gender and Sleep duration related?

Null hypothesis (H_0) – There is no association between gender and sleep duration.

Alternative hypothesis (H_1) – There is an association with gender and sleep duration.

	5-8h	Less than 4h	More than 8h	Total
Female	11	2	1	14
Male	13	1	2	16
Total	24	3	3	30

Table 4.1.18: contingency table: Gender / Sleep Duration

```

> summary(contingency_table)
Number of cases in table: 30
Number of factors: 2
Test for independence of all factors:
  Chisq = 0.7031, df = 2, p-value = 0.7036
  Chi-squared approximation may be incorrect
> result1

Pearson's Chi-squared test

data: contingency_table
X-squared = 0.70312, df = 2, p-value = 0.7036
>

```

Figure 4.1.20: Chi squared sleep/gender.

The data obtained from the t-test indicates,

- Chi squared value = 0.70312
- P-value = 0.7036
- Degree of freedom = 2
- Critical value = 4.605

Since the level of significance is 0.1 p-value obtained from the test is greater than the level of significance.

Critical value > Chi squared value = 4.605 > 0.70312

p- value > Level of significance = 0.7036 < 0.1

H₀ is accepted.

Conclusion – there is no association between gender and sleeping duration.

04. Are Gender and Weekly exercise frequency related?

Null hypothesis (H₀) – There is no association between gender and Weekly exercise frequency.

Alternative hypothesis (H₁) – There is an association with gender and Weekly exercise frequency.

	2-4 times	More than 4 times	Never	Once a week	Rarely	Total
Female	3	0	3	0	8	14
Male	4	2	2	1	7	16
Total	7	2	5	1	15	30

Table 4.1.19: contingency table: Gender / Weekly exercise frequency

```

> summary(contengency_table)
Number of cases in table: 30
Number of factors: 2
Test for independence of all factors:
  Chisq = 3.291, df = 4, p-value = 0.5104
  Chi-squared approximation may be incorrect
> result1

Pearson's Chi-squared test

data:  contengency_table
X-squared = 3.2908, df = 4, p-value = 0.5104
> |

```

Figure 4.1.21: Chi squared gender/weekly exercise frequency.

The data obtained from the t-test indicates,

- Chi squared value = 3.2908
- P-value = 0.5104
- Degree of freedom = 4
- Critical value = 7.779

Since the level of significance is 0.1 p-value obtained from the test is greater than the level of significance.

Critical value > Chi squared value = 7.779 > 3.2908 p-

value > Level of significance = 0.5104 < 0.1

H₀ is accepted.

Conclusion – there is no association between gender and Weekly exercise frequency.

05. Are Academic performance on mental health and Mental health on academic performance related?

Null hypothesis (H₀) – There is no association between Academic performance on mental health and Mental health on academic performance.

Alternative hypothesis (H₁) – There is an association between Academic performance on mental health and Mental health on academic performance.

	No	Not sure	Sometimes	Yes	Total
No	5	0	0	2	7
Sometimes	0	0	8	3	11

Yes	1	1	0	10	12
Not sure	0	0	0	0	0
Total	6	1	8	15	30

Table 4.1.20: contingency table: Mental health and academic performance.

```
> summary(contingency_table)
Number of cases in table: 30
Number of factors: 2
Test for independence of all factors:
  Chisq = 32.04, df = 6, p-value = 1.605e-05
Chi-squared approximation may be incorrect
> result1

Pearson's Chi-squared test

data:  contingency_table
X-squared = 32.038, df = 6, p-value = 1.605e-05
> |
```

Figure 4.1.22: Chi squared mental health and academic performance.

The data obtained from the t-test indicates,

- Chi squared value = 32.038
- P-value = 0.00001605.
- Degree of freedom = 6
- Critical value = 10.645

Since the level of significance is 0.1 p-value obtained from the test is less than the level of significance.

Critical value < Chi squared value = 10.645 < 32.038 p-

value < Level of significance = 0.0001605 < 0.1 H₀ is

rejected.

Conclusion – there is an association between Academic performance on mental health and Mental health on academic performance.

06. Are Gender and Activity levels related?

Null hypothesis (H₀) – There is no association between Gender and Activity level.

Alternative hypothesis (H₁) – There is an association between Gender and Activity level.

	Active	Less active	Not Active	Very active	Total
Female	4	8	1	1	14
male	8	7	0	1	16
Total	12	15	1	2	30

Table 4.1.21: contingency table: Gender and Activity level.

```
> summary(contengency_table)
Number of cases in table: 30
Number of factors: 2
Test for independence of all factors:
  Chisq = 2.2768, df = 3, p-value = 0.517
Chi-squared approximation may be incorrect
> result1

Pearson's Chi-squared test

data:  contengency_table
X-squared = 2.2768, df = 3, p-value = 0.517
> |
```

Figure 4.1.23: Chi squared gender and activity level.

The data obtained from the t-test indicates,

- Chi squared value = 2.2768
- P-value = 0.517.
- Degree of freedom = 3
- Critical value = 6.251

Since the level of significance is 0.1 p-value obtained from the test is greater than the level of significance.

Critical value > Chi squared value = 6.251 > 2.2768

p- value > Level of significance = 0.517 > 0.1

H₀ is accepted.

Conclusion – there is no association between gender and activity level.

Out of all the chi squared tests carried out only mental health on academic performance and academic performance on mental health had an association.

Chapter 5 – Discussion and Conclusions

Out of all current Sliit City Uni students from foundation to third year, 30 students had participated in this survey “Comprehensive mental and physical well-being.”. Out of those 30 students there were 2 more male students than the number of female students. 20 of all the students participated in the survey were from 2nd Year. The majority of the students belonged to the age group 18-24.

Physical health

Out of the 15 attributes used, 8 attributes can be used to conclude on physical health of Sliit City Uni students. The attributes are Sleep duration, height, weight, water intake, activity level, unhealthy habits, weekly exercise frequency, and BMI. The majority of students get the recommended amount of sleep per day according to the data analyzed. Considering the conclusions achieved by hypothesis testing it is required to point out that the majority of students of Sliit City Uni, both male and female fall under the category of “Underweight”. Being underweight can lead to many serious health issues such as Anemia, and Osteoporosis. Students at Sliit City Uni regularly drink enough water as the analyzed data proves. Activity levels and weekly exercise frequency of students were on the negative end. Most students do not engage in exercises weekly and they also tend to be less active throughout the day. Furthermore, the majority of the students have unhealthy habits that can affect them and their physical health negatively.

By considering the above-mentioned data it is possible to conclude that students at Sliit City Uni are slightly below the physically “Healthy” range. It is important to note that it is considered as “slightly” below.

Mental Health

Attributes Sleep duration, Academic performance on mental health, Mental health on academic performance, Unhealthy habits, Mental exhaustion, Self-feeling were used to gather information on mental health level of Sliit City Uni students in the survey. A significant majority of students state that their mental health affects their academic performance negatively and their academic performance also affects their mental health negatively. It is important to note that these 2 attributes are associated. It means students whose mental health is negatively affected by their academic performance tend to also have their academic performance negatively affected by their mental health. Furthermore, students also state that they often feel mentally exhausted while they tend to have a neutral feeling about themselves (both positive and negative)

Considering the factors above-mentioned mental health levels of Sliit City Uni students can be also concluded as below the healthy range. It can be considered as slightly below their physical health level considering the data collected through the survey.

To conclude on overall mental and physical health of Sliit City Uni Students, it is necessary to combine conclusions on both physical and mental health. The conclusions on both physical health and mental health turns out to be negative, indicating that overall level physical and mental well-being of Sliit City Uni students is slightly below than where that level should be. To conclude, overall comprehensive mental and physical health level of Sliit City Uni students is lower than the healthy range. Hence, students should pay more attention to their physical and mental well-being than the amount of attention they are paying now.

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Appendices

R studio commands (Repetitive commands are not presented)

```
#hypothesis testing
install.packages("dplyr")
library(dplyr)
orderbmi <- sort(subset(weightData$BMI, weightData$Gender == "Female"))
resultFemale <- t.test(orderbmi, mu=18.5, conf.level = 0.9, alternative = 'less')
resultFemale
```

Figure 5.1: hypothesis t test

```
Survey_data <- read_excel("withAges.xlsx")
AgeNumerical <- c(Survey_data$Age)
AgeNumerical
sortedAge <- sort(AgeNumerical)
sortedAge
print(sortedAge)
mean(sortedAge)
median(sortedAge)
range(sortedAge)
28-17
var(sortedAge)
sd(sortedAge)
2.809876*2.809876

CustomMode <- function(data) {
  unique <- unique(data)
  mode <- unique[which.max(tabulate(match(data, unique)))]
  return(mode)
}
```

Figure 5.2: numerical analysis

```
YearOfStudy <- table(Survey_data$Year)
YearOfStudy
Yearlbls = c("1st year", "2nd year", "3rd year", "Foundation")
pct <- round(YearOfStudy/sum(YearOfStudy)*100)
lbls <- paste(Yearlbls, pct)
lbls <- paste(lbls, "%", sep="")
pie(YearOfStudy, labels = lbls, col=rainbow(length(lbls)))
```

Figure 5.3: categorical analysis

```
contengency_table <- table(Survey_data$Gender,Survey_data$Sleep)
result1 = chisq.test(contengency_table)
summary(contengency_table)

contengency_table <- table(Survey_data$Gender,Survey_data$Weekly_exercise)
result1 = chisq.test(contengency_table)
summary(contengency_table)
```

Figure 5.4: Chi squared test.